Predicting Fraudulent Transactions Using Machine Learning

Data Analysis Report

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Abstract:

Fraudulent transactions represent a significant threat to the financial ecosystem, causing substantial monetary losses. This study evaluates the effectiveness of Logistic Regression for detecting fraudulent transactions, using a Kaggle dataset that includes anonymized credit card transaction records. The model's performance was evaluated using metrics: accuracy, precision, recall, F1-score, and AUC-ROC. Our results show that Logistic Regression offers reasonable accuracy, though improvements can be made in recall and precision to enhance fraud detection.

Introduction:

Fraudulent transactions have become increasingly frequent, causing significant financial losses. To mitigate these losses Machine learning (ML) and deep learning (DL) techniques have proven effective.

2.1 Overview of Fraudulent Transactions:

Fraudulent transactions involve unauthorized activities within financial systems, credit card fraud, identity theft, and unauthorized account access, leading to financial losses globally.

2.2 Importance of Fraud Detection:

Effective fraud detection minimizes monetary losses, protects customers, and ensures the integrity of financial institutions. Given the increasing sophistication of fraud, automated systems leveraging machine learning models, Logistic Regression, can significantly improve detection efficiency.

2.3 Machine Learning in Fraud Detection:

Logistic Regression is a statistical method used for binary classification. It estimates the probability of an instance belonging to a particular class (fraud or non-fraud) based on the relationship between input features and the outcome. In fraud detection, Logistic Regression helps to identify patterns that differentiate fraudulent transactions from genuine ones.

2.4 Objective of the Research:

The objective is to evaluate Logistic Regression's effectiveness in detecting fraudulent transactions, using accuracy, precision, recall, F1-score, and AUC-ROC as performance metrics.

Methodology:

3.1 Data Collection and Preprocessing:

Dataset Source: Kaggle dataset containing anonymized transactions with features.
Preprocessing Steps:

- **Handling Imbalance:** The dataset is imbalanced, with fraud accounting for a small percentage of transactions. Hot-encoding and splitting data into targets was applied balance the dataset.
- **Standardization:** Features were standardized to ensure comparability.
- **Data Splitting:** The dataset was split into 70% training and 30% testing sets.
- **Null Value Handling:** Any missing values were removed.

3.2 Model Development and Training:

Model Used: Logistic Regression.
Training Process:

- The model was trained on the training set using LogisticRegression from scikit-learn.
- The model was evaluated on both the training and test datasets to ensure generalization.

3.3 Model Evaluation and Optimization:

Evaluation Metrics:

- **Accuracy:** Overall correctness of the model.
- **Precision:** Fraction of predicted fraudulent transactions that were actually fraudulent.
- **Recall:** Fraction of actual fraudulent transactions correctly identified by the model.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC-ROC:** Measures the model's ability to distinguish between fraudulent and non-fraudulent transactions.

Results:

4.1 Descriptive Statistics:

- **Number of Transactions:** 5,090,096.
- **Number of Fraudulent Transactions:** 6,593 (0.13%).
- **Average Transaction Amount:** $88.35.
- **Maximum Transaction Amount:** $25,691.16.

4.2 Model Performance Metrics:

Metric	Logistic Regression
Accuracy	91.09%
Precision	78%
Recall	0.44
F1-Score	0.56
AUC-ROC	0.94

- **Train Accuracy:** 91.09%
- **Test Accuracy:** 91.13%
-

4.3 Confusion Matrix:

Confusion Matrix (Train):

[[5082702   801]
 [3702   2891]]

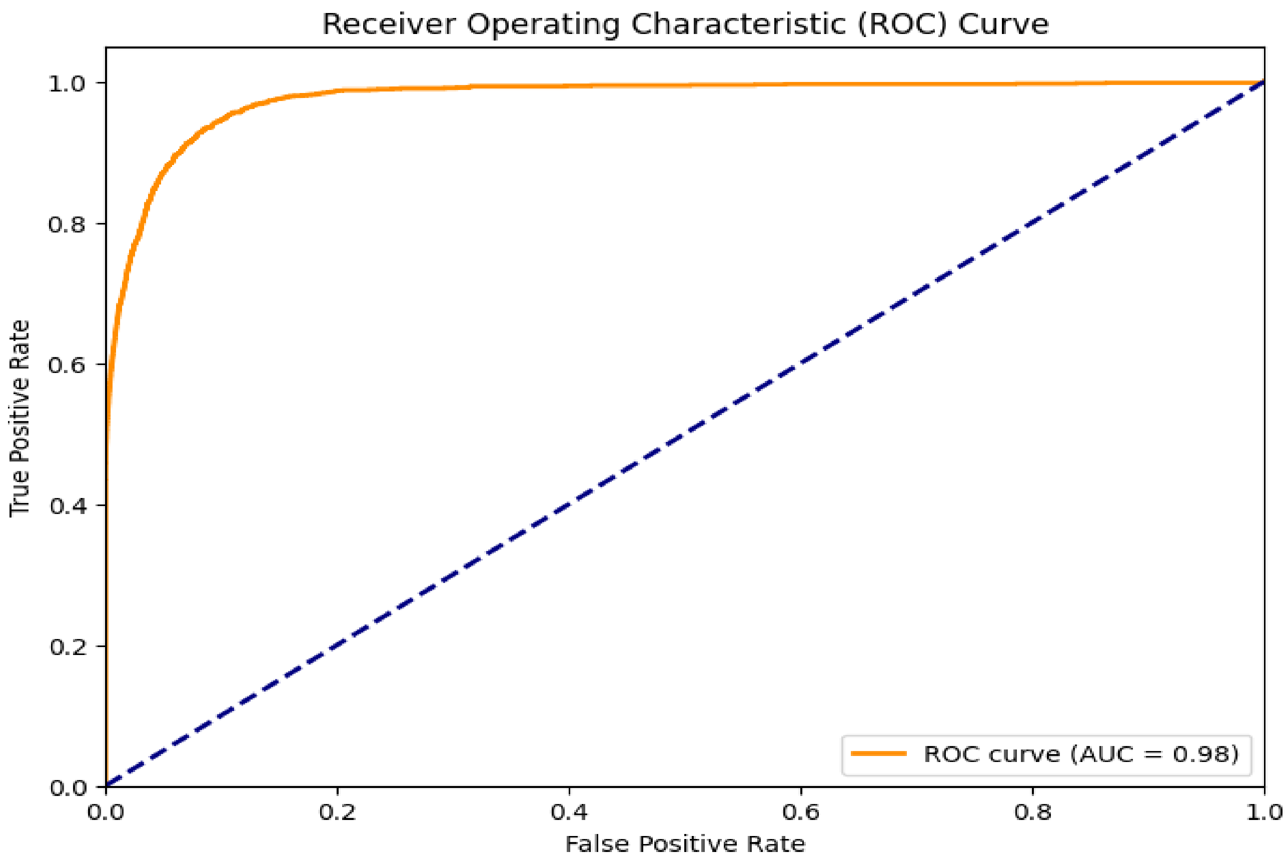
Confusion Matrix (Test):

[[1270709   195]
[924   696]]

- The confusion matrix shows that while the model is very accurate, it has a lower recall, meaning it misses some fraudulent transactions.

4.4 AUC-ROC:

- **AUC-ROC Value:** 0.94, indicating good model performance in distinguishing between fraudulent and non-fraudulent transactions.
- The **ROC Curve** is plotted below to visualize model performance.



Discussion:

5.1 Analysis of Model Performance:

- The **accuracy** of 91.09% (train) and 91.13% (test) shows that the model is quite accurate. However, the low **recall** of 0.44 on the test set suggests the model is not detecting enough fraudulent transactions, which is crucial in fraud detection applications.
- The **AUC-ROC** of 0.94 indicates strong discriminatory power, as it approaches 1.0.

5.2 Insights from Results:

- The **precision** (0.78) and **recall** (0.44) show that while the model is good at predicting non-fraudulent transactions, it fails to identify many fraudulent transactions.
- The **AUC-ROC** value of 0.94 suggests that the model has a good ability to rank transactions by their likelihood of being fraudulent.

5.3 Limitations and Future Improvements:

Limitations:

- The class imbalance might limit the model's ability to detect fraudulent transactions.
- The model's lower recall means many fraudulent transactions are not detected.

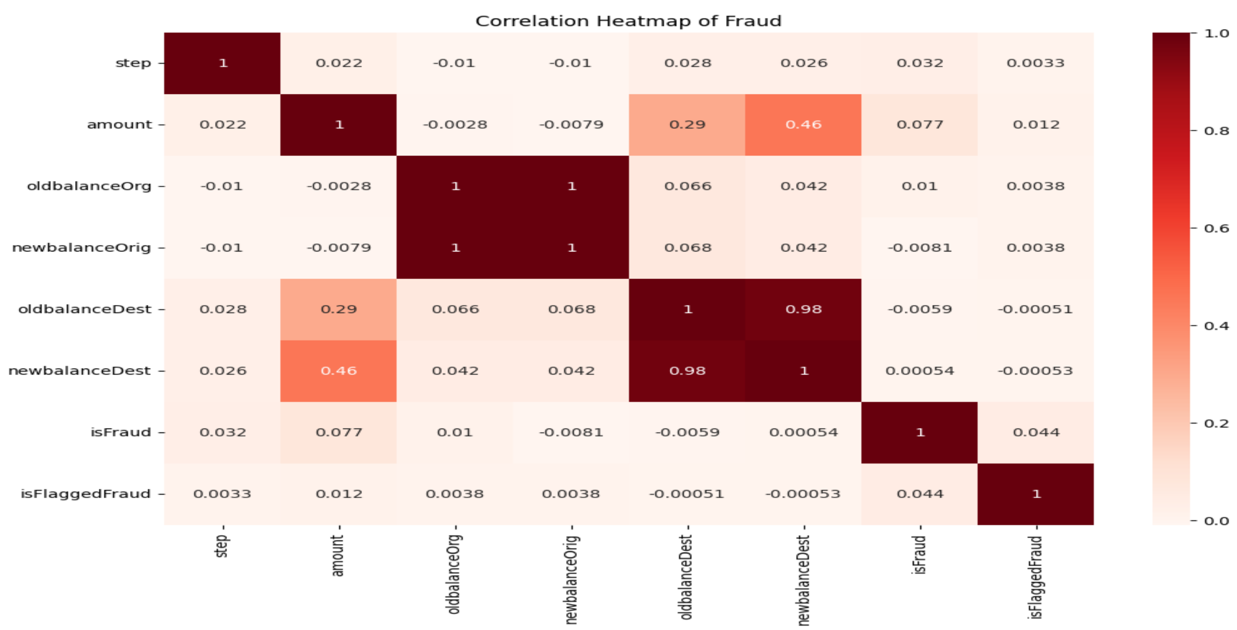
Future Improvements:

- Implement more advanced techniques like **Random Forest** or **Gradient Boosting** for better handling of class imbalance.
- Incorporate **real-time data** for continuous learning and fraud detection in real-time transactions.
- Explore **deep learning** models to capture more complex patterns.

Essential Information:

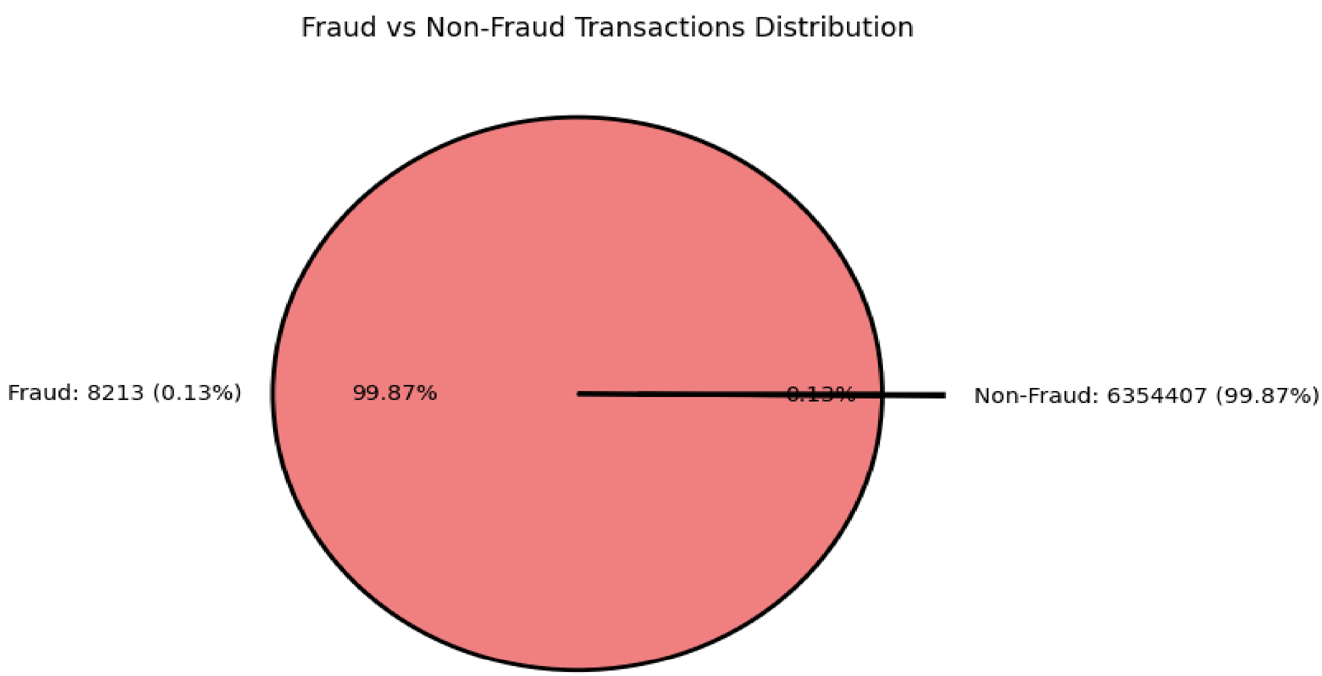
7.1 Correlation Matrix:

A heatmap revealed significant multicollinearity between variables such as "oldbalanceOrg" and "newbalanceOrig" (correlation = 1), as well as "oldbalanceDest" and "newbalanceDest" (correlation = 0.98), indicating potential redundancy in the dataset. Most other features, including "amount," show weak correlations with the target variable "isFraud," suggesting that fraud detection may require a multivariate approach.



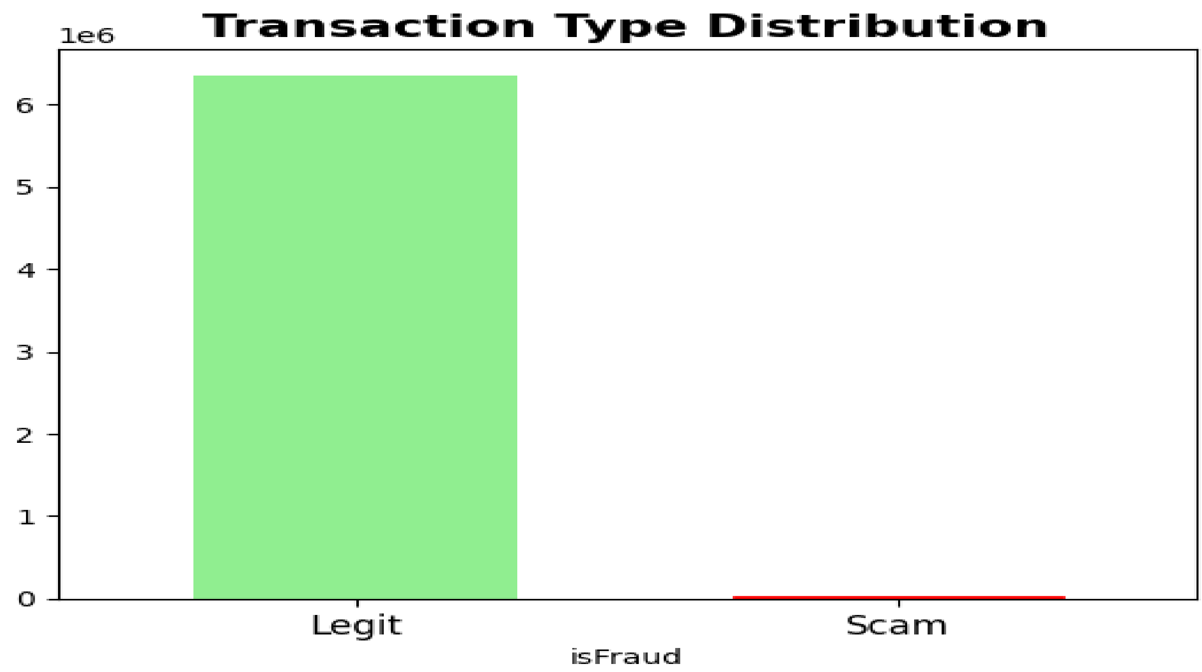
7.2 Pie Plot:

The pie chart visually reinforces that fraudulent transactions are extremely rare (0.13%) compared to legitimate ones (99.87%), emphasizing the need for specialized techniques to address the class imbalance in machine learning models.



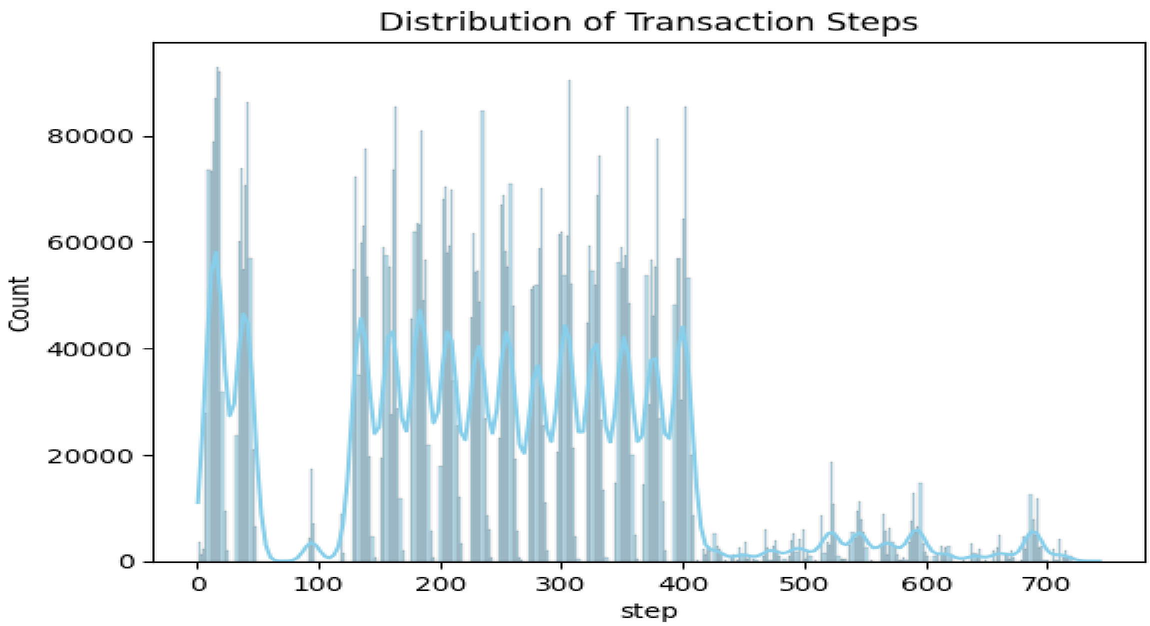
7.3 Bar Plot:

The bar plot shows a significant imbalance between legitimate (Legit) and fraudulent (Scam) transactions, with legitimate transactions far outnumbering fraudulent ones. This highlights the challenge of class imbalance in fraud detection.



7.4 Histplot

The histogram implies that most transactions occur within early steps or time intervals, with periodic peaks indicating possible batch processing or regular system events. After step 400, transaction frequency drops significantly. This suggests that the majority of activity happens in a specific time window, which could be important for monitoring or investigating anomalies, such as fraud, and understanding system behavior.



Conclusion:

Logistic Regression demonstrates a solid performance in detecting fraudulent transactions with an accuracy of 91.09%. However, the model's lower recall and the class imbalance issue suggest room for improvement. Future models should address these limitations, possibly with more advanced techniques and real-time processing for enhanced fraud detection.

References:

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